

Chapter 62

ADC Controller (ADC)

62.1 Overview

ESP32-P4 integrates two 12-bit successive approximation ADCs (SAR ADCs) for measuring analog signals from up to 14 pins. Figure 62.1-1 shows the basic architecture of the SAR ADC module on ESP32-P4.

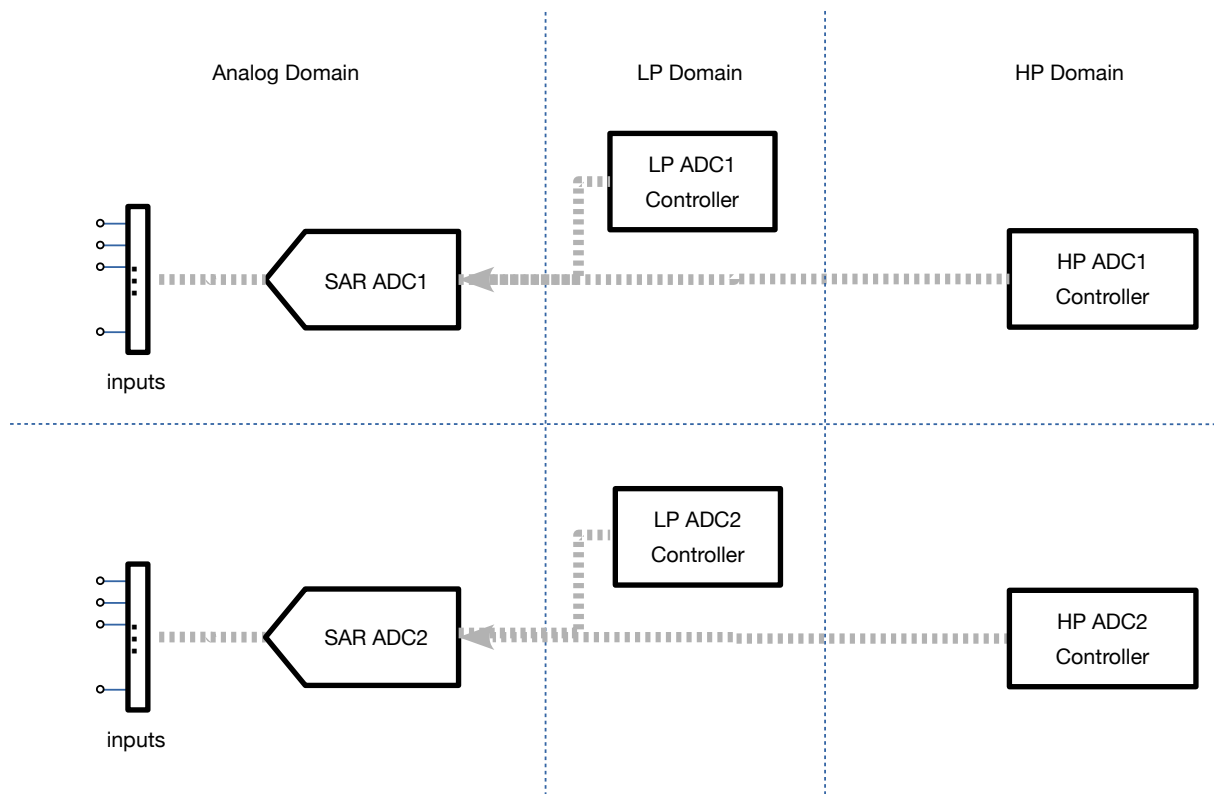


Figure 62.1-1. SAR ADC Simplified Architecture

As the figure shows, the SAR ADCs are managed by four dedicated controllers:

- Two HP (high-performance) controllers: HP ADC1 Controller and HP ADC2 Controller, designed for multi-channel sampling mode, supporting continuous transfer of conversion results to memory via the GDMA interface.
- Two LP (low-power) controllers: LP ADC1 Controller and LP ADC2 Controller, supporting operation in sleep mode and one-shot sampling mode.

62.2 Terminology

The following terms related to SAR ADC are defined in the context of the *ESP32-P4 Technical Reference Manual* to help readers better understand this document:

SAR ADC:	Including analog ADC circuit and digital ADC controller. The HP ADC and LP ADC in the following text share the analog ADC circuit, but have their own controllers, namely the HP ADC controllers and LP ADC controllers. The SAR ADC mentioned in the text refers to both HP ADC and LP ADC.
HP ADC:	High-performance ADC, supporting multi-channel sampling, GDMA data transfer, etc., not available in sleep mode.
LP ADC:	Low-power ADC, supporting one-shot sampling mode, can be used in sleep mode.
One-shot sampling mode:	In this mode, LP ADC samples one channel at a time.
Multi-channel sampling mode:	In this mode, HP ADC sequentially samples a group of channels or continuously samples a single channel.
Dual HP ADC sampling:	It means two HP ADCs sample either simultaneously or alternatively.
Conversion result:	The conversion result is the binary digital value obtained after the analog-to-digital conversion process. It is sometimes written as conversion data.
Filtered data:	Filtered data is the result of processing conversion data through a filter.
Sampling:	The term typically refers to the entire process of sampling analog inputs, converting the sampled data, and transferring the conversion results to memory. In certain contexts and stages of the process, sampling may also refer to the SAR ADC capturing data points from an analog signal.

62.3 Features

The SAR ADCs have the following features:

- Support HP ADC_x controllers and LP ADC_x controllers to get control of SAR ADC_x via software, _x = 1, 2
- 12-bit resolution
- Analog inputs sampling from up to 14 pins
- HP ADC_x controllers:
 - Provide multi-channel sampling control module, supporting multi-channel sampling mode, with configurable channel sampling sequence
 - Provide mode control module, supporting dual HP ADC sampling
 - Provide two filters with configurable filter coefficients

- Provide two threshold monitors that can trigger an interrupt when the filtered data is above a high threshold or below a low threshold
- Support continuous transfer of conversion results to memory via the GDMA interface
- LP ADC_x controllers:
 - Support one-shot sampling mode
 - Support sampling in sleep mode (e.g., Deep-sleep)
- Support for several Event Task Matrix (ETM) related events and tasks

62.4 Architecture

The major components of SAR ADCs and their interconnections are shown in Figure 62.4-1.

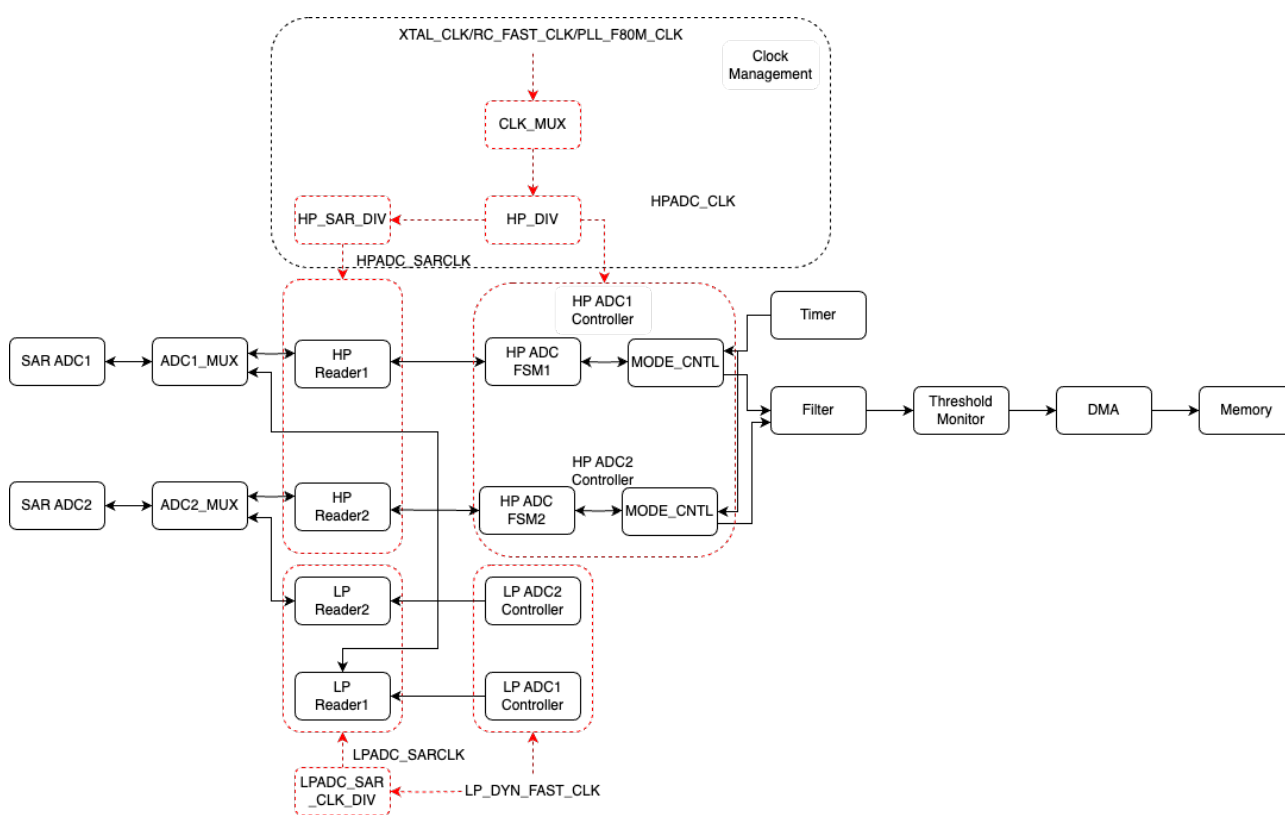


Figure 62.4-1. SAR ADC Architecture

- →: clock signals
- : clock divider, clock mux, and the blocks the clock works for

As Figure 62.4-1 shows, the SAR ADC module contains the following major functional blocks:

- SAR ADC1: Samples analog inputs from up to 8 pins
- SAR ADC2: Samples analog inputs from up to 6 pins
- Clock Management: Selects clock source and division
- Timer: Dedicated timer for the HP ADC_x controllers to generate sampling enable signal.
- HP ADC FSM_x: FSM1 and FSM2 that can:

- receive the sampling enable signal from the timer.
 - determine the conversion rules defined in the patterns for HP ADC_x.
 - drive the HP Reader_x module to read conversion data.
 - transfer conversion data to the filter.
- Filter: Filter0 and filter1, used for filtering conversion results in multi-channel sampling mode.
 - Threshold Monitor: Threshold monitor 0 and threshold monitor 1 that trigger an interrupt when the filtered data is above a high threshold or below a low threshold.
 - MODE_CNTL: Used for filtering the sampling signal triggered by the timer, supporting dual HP ADC sampling mode.
 - HP Reader_x: HP reader 1 and HP reader 2, driven by HP ADC FSM_x to read data from HP ADC_x.
 - LP Reader_x: LP reader 1 and LP reader 2, driven by LP ADC_x Controller to read data from LP ADC_x.
 - LP ADC_x Controller: LP ADC1 Controller and LP ADC2 Controller. The controllers provide sampling enable signal, drive the LP Reader_x module to read the conversion results of LP ADC_x, and store the conversion results.

62.5 Functional Description

62.5.1 SAR ADC Power Up

The SAR ADCs can be powered up by setting [PMU_XPD_PERIF_I2C](#) and [PMU_PERIF_I2C_RST](#). The ADC sampling can be conducted immediately after power-up. Users need not be concerned about wait time, as it has been accounted for in the hardware design.

62.5.2 SAR ADC Channels

The SAR ADCs have 14 channels that are connected to 14 pins on the chip. To sample an analog signal, the SAR ADCs must first select the analog pin to measure via an internal multiplexer.

Table [62.5-1](#) shows the pins used as SAR ADC channels.

Table 62.5-1. SAR ADC Channels and Chip Pins

Chip Pins	SAR ADC Channel	SAR ADC Selection
GPI016	0	SAR ADC1
GPI017	1	
GPI018	2	
GPI019	3	
GPI020	4	
GPI021	5	
GPI022	6	
GPI023	7	
GPI049	0	
GPI050	1	

SAR ADC2

Chip Pins	SAR ADC Channel	SAR ADC Selection
GPIO51	2	
GPIO52	3	
GPIO53	4	
GPIO54	5	

62.5.3 SAR ADC Clock

The clock for the HP ADC_x Controller has three possible sources, selected by [HP_SYS_CLKRST_ADC_CLK_SRC_SEL](#):

- XTAL_CLK
- RC_FAST_CLK
- PLL_F80M_CLK

The following clocks can be divided from the HP ADC_x Controller's clock:

- HPADC_SARCLK: It is the operating clock of HP ADC_x and HP Reader_x.
- HPADC_CLK: It is the operating clock of HP ADC FSM_x.

The frequency of HPADC_SARCLK can affect the sampling accuracy. If the HPADC_SARCLK frequency is higher than 5 MHz, the sampling accuracy will decrease. HPADC_SARCLK is divided from HPADC_CLK via a dedicated divider. The division is set in [ADC_SAR_CLK_DIV](#).

It takes 25 HPADC_SARCLK clock cycles to complete one sampling, so the maximum sampling rate is limited by the frequency of HPADC_SARCLK.

The LP ADC_x Controller is clocked from LP_DYN_FAST_CLK.

The following clocks can be divided from the LP ADC_x Controller's clock:

- LPADC_SARCLK: It is the operating clock of LP ADC_x and LP Reader_x. The division set in [LPADC_SAR1_CLK_DIV](#) and [LPADC_SAR2_CLK_DIV](#) should be at least 2. The frequency of LPADC_SARCLK should not exceed 5 MHz.
- LPADC_CLK: It is the operating clock of LP ADC FSM_x.

For more information about clocks, please refer to Chapter 10 [Reset and Clock](#).

62.5.4 SAR ADC Conversion and Attenuation

The SAR ADCs can measure analog voltages from 0 mV to V_{ref} . V_{ref} is the SAR ADCs' internal reference voltage. The conversion result (*data*) is a 12-bit digital value, which is the raw data. To calculate the voltage V_{data} based on the raw data, the following formula can be used:

$$V_{data} = \frac{V_{ref}}{4095} \times data$$

k is the coefficient corresponding to the configured attenuation.

To convert voltages larger than V_{ref} , apply attenuation to the input signals. The supported attenuation levels are:

- 0 dB ($k \approx 100\%$)
- 2.5 dB ($k \approx 75\%$)
- 6 dB ($k \approx 50\%$)
- 12 dB ($k \approx 25\%$)

62.5.5 HP ADC FSM

In multi-channel sampling mode HP ADC FSM (hereinafter referred to as FSM) generates all types of signals used in the sampling process. Figure 62.5-1 illustrates how the FSM works.

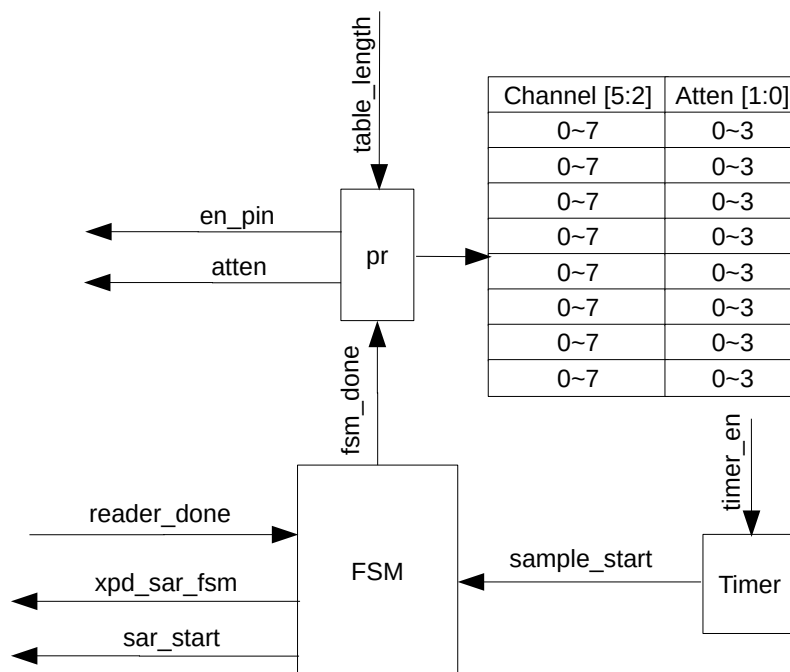


Figure 62.5-1. HP ADC FSM Block Diagram

Wherein:

- Timer: a dedicated timer for the HP ADC controller to generate **sample_start** signal.
- pr: a pointer to the pattern table that defines the conversion rules for HP ADC. FSM sends out corresponding signals based on the conversion rules.

Set [ADC_TIMER_EN](#) to enable the timer. The timeout event of this timer triggers a **sample_start** signal. This signal drives the FSM module to start sampling. When the FSM module receives the **sample_start** signal, it starts the following operations:

- Powers up HP ADC.
- Determines the conversion rules (selected sample channels and attenuations) defined in the patterns that the current pr points to.
- Outputs the **en_pin** and **atten** signals corresponding to the conversion rules to the analog side.
- Initiates the **sar_start** signal and starts sampling.

When FSM receives the `reader_done` signal from HP Reader, it starts the following operations:

- Stops sampling.
- Transfers the conversion result to the filter. Then the threshold monitor transfers the filtered result to memory via GDMA (see Figure 62.4-1).
- Updates `pr` and waits for the next sampling. The pointer `pr` counts cyclically between 0 and `ADC_SAR1/2_PATT_LEN` (`table_length`).

62.5.6 HP ADC Pattern Table

HP ADC FSM1/2 each contain a pattern table configured by `ADC_SARx_PATT_TABm_REG`, with *m* indicating registers 1 ~ 4. Each register contains four patterns and each pattern is 6 bits wide, as below shows.

(reserved)								cmd0				cmd1				cmd2				cmd3			
3124								2318				1712				116				50			
00000000								0x0000				0x0000				0x0000				0x0000			

`cmd n` (*n* = 0 - 3) represents patterns 0 ~ 3.

Figure 62.5-2. `ADC_SARx_PATT_TAB1_REG` Contains Patterns 0 - 3

(reserved)																cmd4				cmd5				cmd6				cmd7																			
31								24								23				18				17				12				11				6				5				0			
0 0 0 0 0 0 0 0								0x0000								0x0000				0x0000				0x0000				0x0000																			

`cmd n` (*n* = 4 - 7) represents patterns 4 ~ 7.

Figure 62.5-3. `ADC_SARx_PATT_TAB2_REG` Contains Patterns 4 - 7

(reserved)								cmd8				cmd9				cmd10				cmd11			
31		24		23		18		17		12		11		6		5		0					
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				

`cmd n` (*n* = 8 - 11) represents patterns 8 ~ 11.

Figure 62.5-4. `ADC_SARx_PATT_TAB3_REG` Contains Patterns 8 - 11

(reserved)								cmd12				cmd13				cmd14				cmd15																								
31								24	23								18	17								12	11								6	5								0
0	0	0	0	0	0	0	0	0	0x0000				0x0000				0x0000				0x0000				0x0000																			

`cmd n` (*n* = 12 - 15) represents patterns 12 ~ 15.

Figure 62.5-5. `ADC_SARx_PATT_TAB4_REG` Contains Patterns 12 - 15

Each pattern is 6 bits wide, consisting of two fields where conversion rules are stored. Figure 62.5-6 shows the pattern format and is followed by the descriptions of each field.

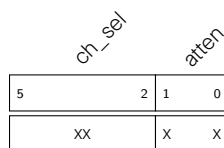


Figure 62.5-6. Pattern Structure

atten Configures attenuation:

- 0: 0 dB
- 1: 2.5 dB
- 2: 6 dB
- 3: 12 dB

ch_sel Configures channel. For HP ADC1, values 0-7 correspond to channels 0-7 individually, for HP ADC2, values 0-5 correspond to channels 0-5 individually.

62.5.7 HP ADC Pattern Configuration Example for Multi-channel Sampling

Assuming you want to implement the following sequence of HP ADC1 multi-channel sampling:

- Sample channel 2 of HP ADC1 with attenuation of 12 dB;
- Sample channel 0 of HP ADC1 with attenuation of 2.5 dB.

Then, the detailed configuration is as follows:

- Configure the first pattern (cmd0) of `ADC_SAR1_PATT_TAB1_REG[5:0]`:

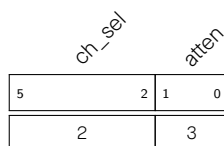


Figure 62.5-7. cmd0 configuration

atten write 3 to this field, to set the attenuation to 12 dB.

ch_sel write 2 to this field, to select channel 2.

- Configure the second pattern (cmd1) of `ADC_SAR1_PATT_TAB1_REG[11:6]`:

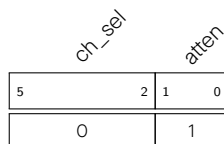


Figure 62.5-8. cmd1 Configuration

atten write 1 to this field, to set the attenuation to 2.5 dB.

ch_sel write 0 to this field, to select channel 0.

- Configure `ADC_SAR1_PATT_LEN` to 1, i.e., set pattern table size to (this value + 1) = 2. Then patterns cmd0 and cmd1 will be used.

- Enable the timer so that the HP ADC1 Controller starts sampling the two channels periodically.

62.5.8 Dual HP ADC Sampling Control

`ADC_WORK_MODE` controls how two HP ADCs work together:

- 0: Single HP ADC sampling, i.e., only one HP ADC samples. When a sampling request is triggered by the timer, `ADC_SAR_SEL` controls which FSM receives the sampling request:
 - 0: HP ADC FSM1, i.e., using HP ADC1 for sampling.
 - 1: HP ADC FSM2, i.e., using HP ADC2 for sampling.
- 1: Dual HP ADC simultaneous sampling. When a sampling request is triggered by the timer, both HP ADC FSM1 and HP ADC FSM2 receive the sampling request.
- 2: Dual HP ADC alternative sampling. When an odd number of sampling requests are triggered by the timer, HP ADC FSM1 receives the sampling request. When an even number of sampling requests are triggered by the timer, HP ADC FSM2 receives the sampling request.

62.5.9 HP ADC Filters

The HP ADC_x controllers provide two filters for filtering conversion results in multi-channel sampling mode. Both filters can be configured to any HP ADC channel, but cannot be configured to the same channel. If the two filters are configured to the same channel, the first one takes effect.

The filtered data is determined by the following equation:

$$data_{cur} = \frac{(k-1)data_{prev}}{k} + \frac{data_{in}}{k} + 0.5$$

- $data_{cur}$: the filtered data
- $data_{in}$: the conversion result
- $data_{prev}$: the last filtered data
- k : the filter coefficient

The filters are configured as follows:

- Configure `ADC_FILTER_CHANNELx` to select the HP ADC channel for filter x , $x=0, 1$.
- Configure `ADC_FILTER_FACTORx` to set the coefficient k for filter x , $x=0, 1$.

62.5.10 HP ADC Threshold Monitors

The HP ADC_x controllers provide two threshold monitors to monitor the filtered data in multi-channel sampling mode. When the data is above the high threshold, a high threshold interrupt is triggered; when the data is below the low threshold, a low threshold interrupt is triggered. Both monitors can be configured to any HP ADC channel, but cannot be configured to the same channel.

Threshold monitors are configured as follows:

- Set `ADC_THRESx_EN` to enable threshold monitor x ($x=0, 1$).
- Configure `ADC_THRESx_LOW` to set a low threshold.

- Configure `ADC_THRESx_HIGH` to set a high threshold.
- Configure `ADC_THRESx_CHANNEL` to select the channel to monitor.

62.5.11 HP ADC GDMA Support

When HP ADC converts data on multiple channels, it is necessary to enable GDMA so that the data can be transferred to memory continuously.

GDMA support is triggered by the timer in the HP ADC_x controllers. Users can switch the GDMA data path to the HP ADC controller by configuring `ADC_APB_ADC_TRANS`. For specific GDMA configuration, please refer to Chapter 4 *GDMA Controller (GDMA-AHB, GDMA-AXI)*.

The HP ADC eventually passes 32-bit data to GDMA. The data format is shown in Figure 62.5-9.

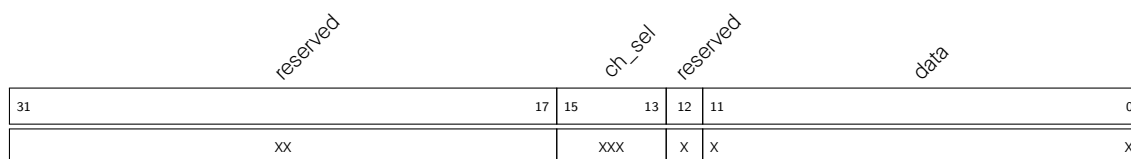


Figure 62.5-9. GDMA Data Format

data 12-bit HP ADC conversion result

ch_sel 3-bit channel information

62.5.12 LP ADC Wake-Up Modes for Automatic Monitoring

The LP ADC_x controllers support automatic monitoring of the conversion data. A dedicated timer generates LP ADC sampling signals periodically based on the value of `LPADC_ADCx_HW_READ_RATE_I`. Then the conversion data is fed into a threshold monitor (not the HP ADC threshold monitors). If the data exceeds the threshold, an interrupt will be generated.

There are two wake-up modes for automatic monitoring, selected by `LPADC_SARx_WAKEUP_MODE`:

- 0: Absolute value mode
Monitors the absolute value of the conversion data. Configure `LPADC_SARx_WAKEUP_TH_LOW` and `LPADC_SARx_WAKEUP_TH_HIGH` to set the thresholds. Wake-up will be triggered if the data is above the high threshold or below the low threshold.
- 1: Change value mode
Monitors the changes of the conversion data. If the increment of two consecutive samplings exceeds the high threshold configured in `LPADC_SARx_WAKEUP_TH_HIGH`, or the decrement of two consecutive samplings exceeds the low threshold configured in `LPADC_SARx_WAKEUP_TH_LOW`, a wake-up will be triggered. For example, when `LPADC_SARx_WAKEUP_TH_LOW` is set to 8, and the two consecutive conversion results are 28 and 19, resulting in a decrement of 9, a wake-up will be triggered.

62.6 Event Task Matrix Feature

The SAR ADC on ESP32-P4 supports the Event Task Matrix (ETM) function, which allows SAR ADC's ETM tasks to be triggered by any peripherals' ETM events, or SAR ADC's ETM events to trigger any peripherals' ETM

tasks. This section introduces the ETM tasks and events related to SAR ADC. For more information, please refer to Chapter [13 Event Task Matrix \(ETM\)](#).

The SAR ADC can receive the following ETM tasks:

- ADC_TASK_SAMPLE0: LP ADC starts one-shot sampling when this task is triggered.
- ADC_TASK_START0: HP ADC starts multi-channel sampling when this task is triggered.
- ADC_TASK_STOP0: SAR ADC stops sampling when this task is triggered.

The SAR ADC can generate the following ETM events:

- ADC_EVT_CONV_CMPLT0: Generated each time SAR ADC completes a sampling in either one-shot sampling mode or multi-channel sampling mode.
- ADC_EVT_EQ_ABOVE_THRESH x : Generated when the HP ADC's filtered data is above the threshold. $x = 0, 1$, representing threshold monitor 0, 1.
- ADC_EVT_EQ_BELOW_THRESH x : Generated when the HP ADC's filtered data is below the threshold. $x = 0, 1$, representing threshold monitor 0, 1.
- ADC_EVT_STARTED0: Generated when SAR ADC begins sampling; one-shot sampling will not trigger this event.
- ADC_EVT_STOPPED0: Generated when SAR ADC stops sampling, one-shot sampling will not trigger this event.

In practical applications, SAR ADC's ETM events can trigger its own ETM tasks. For example, the ADC_EVT_EQ_ABOVE_THRESH x event can trigger the ADC_TASK_STOP0 task.

62.7 Interrupts

ESP32-P4's SAR ADC can generate the following interrupt signals that will be sent to the [Interrupt Matrix](#).

- ADC_INTR
- LP_ADC_INTR

The following internal interrupt sources from SAR ADC can generate the interrupt signal ADC_INTR:

- ADC_SAR x _DONE_INT: Triggered when HP ADC x completes one sampling.
- ADC_THRES x _HIGH_INT: Triggered when the filtered data is above the high threshold.
- ADC_THRES x _LOW_INT: Triggered when the filtered data is below the low threshold.

The following internal interrupt sources from SAR ADC can generate the interrupt signal LP_ADC_INTR:

- LPADC_COCPU_SARADC x _DONE_INT: Triggered when LP ADC x completes one sampling.
- LPADC_COCPU_SARADC x _ERROR_INT: Triggered when sampling error occurs.
- LPADC_COCPU_SARADC x _WAKE_INT: Triggered when LP ADC x generates a wake-up event (such as conversion data exceeding the threshold).

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [62.9 Register Summary](#).

62.8 Programming Procedure

62.8.1 HP ADC Multi-Channel Sampling Mode

The multi-channel sampling mode can be configured with the following procedure:

1. Set [PMU_XPD_PERIF_I2C](#) and [PMU_PERIF_I2C_RST](#) to power up HP ADC.
2. Configure [HP_SYS_CLKRST_ADC_CLK_SRC_SEL](#) to select HP ADC clock source.
3. Configure [HP_SYS_CLKRST_ADC_CLK_DIV_NUM](#) and [HP_SYS_CLKRST_ADC_SARx_CLK_DIV_NUM](#) to set clock division.
4. Set [HP_SYS_CLKRST_ADC_CLK_EN](#) to enable HP ADC clock.
5. Set [LPADC_SARx_DIG_FORCE](#) to enable the sampling channel.
6. Configure the pattern table as described in Section [62.5.6 HP ADC Pattern Table](#).
7. Configure channels and filtering coefficients as described in Section [62.5.9 HP ADC Filters](#).
8. Configure threshold monitoring as described in Section [62.5.10 HP ADC Threshold Monitors](#), as needed.
9. Set [ADC_APB_ADC_TRANS](#) to use GDMA.
10. Configure [ADC_TIMER_TARGET](#) to set trigger target for HP ADC timer.
11. Set [ADC_TIMER_EN](#) to enable the timer.

The timer timeout will trigger HP ADC FSM to start sampling according to the pattern table. The conversion result will be automatically stored in memory. When the sampling reaches the specified limit in [ADC_APB_ADC_EOF_NUM](#), it will terminate.

Once sampling is complete, an [ADC_SARx_DONE_INT_RAW](#) interrupt will be generated.

62.8.2 LP ADC One-shot Sampling Mode

The one-shot sampling mode can be configured with the following procedure:

1. Set [PMU_XPD_PERIF_I2C](#) and [PMU_PERIF_I2C_RSTB](#) to power up LP ADC.
2. Configure [LPPERI_LPADC_FUNC_DIV_NUM](#) and [LPPERI_LPADC_SARx_DIV_NUM](#) to set the frequency of [LPADC_CLK](#) and [LPADCx_SARCLK](#).
3. Configure [LPADC_SARx_EN_PAD](#) and [LPADC_SARx_ATTEN](#) to select sampling channels and attenuation.
4. Set [LPADC_SAR_MEASx_START_SAR](#) to enable the one-shot sampling mode of LP ADCx.

Once sampling is complete, an [LPADC_COCPU_SARADCx_INT_RAW](#) interrupt is generated. Software can read the conversion result from [LPADC_SAR_MEASn_DATA_SAR](#).

62.8.3 LP ADC Automatic Monitoring

The configuration process for the automatic monitoring of the LP ADC controller is as follows:

1. Configure [LPADC_ADCx_HW_READ_RATE_I](#) to set the sampling rate.
2. Configure [LPADC_SARx_WAKEUP_MODE](#) to select the wake-up mode.

3. Configure `LPADC_SARx_WAKEUP_TH_HIGH` and `LPADC_SARx_WAKEUP_TH_LOW` to set the high and low thresholds.
4. Set `LPADC_SARx_WAKEUP_EN` to enable automatic monitoring.
5. Set `LPADC_ADCx_HW_READ_EN_I` to enable periodic monitoring.

During the automatic monitoring process, the conversion data will not be stored, but software can always read the last conversion data from `LPADC_SAR_MEASn_DATA_SAR`.

62.9 Register Summary

62.9.1 HP ADC Register Summary

The addresses in the table below are relative to the base address of HP ADC (written as ADC Controller) in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
HP ADC Configuration Registers			
ADC_CTRL_REG	Configuration register for HP ADC Controller	0x0000	R/W
ADC_CTRL2_REG	Configuration register for HP ADC Controller.	0x0004	R/W
ADC_FILTER_CTRL1_REG	Configuration register 1 for HP ADC filter.	0x0008	R/W
ADC_SAR1_PATT_TAB1_REG	Pattern table register 1 for HP ADC1.	0x0018	R/W
ADC_SAR1_PATT_TAB2_REG	Pattern table register 2 for HP ADC1.	0x001C	R/W
ADC_SAR1_PATT_TAB3_REG	Pattern table register 3 for HP ADC1.	0x0020	R/W
ADC_SAR1_PATT_TAB4_REG	Pattern table register 4 for HP ADC1.	0x0024	R/W
ADC_SAR2_PATT_TAB1_REG	Pattern table register 1 for HP ADC2.	0x0028	R/W
ADC_SAR2_PATT_TAB2_REG	Pattern table register 2 for HP ADC2.	0x002C	R/W
ADC_SAR2_PATT_TAB3_REG	Pattern table register 3 for HP ADC2.	0x0030	R/W
ADC_SAR2_PATT_TAB4_REG	Pattern table register 4 for HP ADC2.	0x0034	R/W
ADC_FILTER_CTRL0_REG	Configuration register 0 for HP ADC filter.	0x003C	R/W
ADC_THRES0_CTRL_REG	Filtered data threshold control register 0.	0x0044	R/W
ADC_THRES1_CTRL_REG	Filtered data threshold control register 1.	0x0048	R/W
ADC_THRES_CTRL_REG	Filter configuration register.	0x004C	R/W
ADC_INT_ENA_REG	Enable register of HP ADC interrupts.	0x0050	R/W
ADC_INT_RAW_REG	Raw register of HP ADC interrupts.	0x0054	R/WTC/SS
ADC_INT_ST_REG	State register of HP ADC interrupts.	0x0058	RO
ADC_INT_CLR_REG	Clear register of HP ADC interrupts.	0x005C	WT
ADC_DMA_CONF_REG	GDMA configuration register for HP ADC.	0x0060	R/W
HP ADC Version Control Registers			
ADC_CTRL_DATE_REG	HP ADC version control register	0x03FC	RO

62.9.2 LP ADC Register Summary

The addresses in the table below are relative to LP ADC's base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
LP ADC Configuration Registers			
LPADC_READER1_CTRL_REG	LP ADC1 Reader control register.	0x0000	R/W

Name	Description	Address	Access
LPADC_MEAS1_CTRL2_REG	LP ADC1 configuration registers.	0x000C	varies
LPADC_MEAS1_MUX_REG	LP or HP ADC controller selection register for SAR ADC1.	0x0010	R/W
LPADC_ATTEN1_REG	LP ADC1 attenuation registers.	0x0014	R/W
LPADC_READER2_CTRL_REG	LP ADC2 Reader control register.	0x0024	R/W
LPADC_MEAS2_CTRL2_REG	LP ADC2 configuration registers.	0x0030	varies
LPADC_MEAS2_MUX_REG	LP or HP controller selection register for SAR ADC2.	0x0034	R/W
LPADC_ATTEN2_REG	LP ADC2 attenuation registers.	0x0038	R/W
LP ADC Power Control Registers			
LPADC_FORCE_WPD_SAR_REG	LP ADC Controllers power control registers.	0x003C	R/W
LP ADC Interrupt Registers			
LPADC_COCPU_INT_RAW_REG	Raw register of LP ADC interrupts.	0x0048	R/WTC/SS
LPADC_INT_ENA_REG	Enable register of LP ADC interrupts.	0x004C	R/WTC
LPADC_INT_ST_REG	Status register of LP ADC interrupts.	0x0050	RO
LPADC_INT_CLR_REG	Clear register of LP ADC interrupts.	0x0054	WT
LPADC_INT_ENA_W1TS_REG	LPADC_INT_ENA_REG configuration register.	0x0058	WT
LPADC_INT_ENA_W1TC_REG	LPADC_INT_ENA_REG configuration register.	0x005C	WT
LP ADC Wake-up Control Registers			
LPADC_WAKEUP1_REG	LP ADC1 wake-up configuration registers.	0x0060	varies
LPADC_WAKEUP2_REG	LP ADC2 wake-up configuration registers.	0x0064	varies
LPADC_WAKEUP_SEL_REG	Wake-up source selection register.	0x0068	R/W
LPADC_SAR1_HW_WAKEUP_REG	LP ADC1 automatic monitoring configuration registers.	0x006C	R/W
LPADC_SAR2_HW_WAKEUP_REG	LP ADC2 automatic monitoring configuration registers.	0x0070	R/W

62.10 Registers

62.10.1 HP ADC Registers

The addresses in the following section are relative to the base address of HP ADC (written as ADC Controller) provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 62.1. ADC_CTRL_REG (0x0000)

(reserved)		ADC_XPD_SAR2_FORCE		ADC_XPD_SAR1_FORCE		(reserved)		ADC_SAR2_PATT_P_CLEAR		ADC_SAR1_PATT_P_CLEAR		ADC_SAR2_PATT_LEN		ADC_SAR1_PATT_LEN		ADC_SAR_CLK_DIV		ADC_SAR_CLK_GATED		ADC_SAR_SEL		ADC_WORK_MODE		reserved	
31	30	29	28	27	26	25	24	23	22	21	18		17	14		13	6		5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	15		15		4		1		0	0	0	0				

ADC_WORK_MODE Configures dual HP ADC sampling.

- 0: Single HP ADC sampling
 - 1: Dual HP ADC simultaneous sampling
 - 2: Dual HP ADC alternative sampling.
- (R/W)

ADC_SAR_SEL Configures which HP ADC to use for single HP ADC sampling.

- 0: Use HP ADC1
 - 1: Use HP ADC2
- (R/W)

ADC_SAR_CLK_GATED Enable HP ADC clock gate when HP ADC is in idle.

- 0: Disable
 - 1: Enable
- (R/W)

ADC_SAR_CLK_DIV Configures HP ADC clock division. Division = configured value + 1. (R/W)

ADC_SAR1_PATT_LEN Configures how many pattern table entries will be used for HP ADC1.

- 0: Use only cmd0
 - 1: Use cmd0 and cmd1
 - n: Use cmd0 to cmdn, the maximum n is 7
- (R/W)

ADC_SAR2_PATT_LEN Configures how many pattern table entries will be used for HP ADC2.

- 0: Use only cmd0
 - 1: Use cmd0 and cmd1
 - n: Use cmd0 to cmdn, the maximum n is 7
- (R/W)

ADC_SAR1_PATT_P_CLEAR Clears the pointer of pattern table for HP ADC1 Controller.

- 0: No effect
 - 1: Clear
- (R/W)

ADC_SAR2_PATT_P_CLEAR Clears the pointer of pattern table for HP ADC2 Controller.

- 0: No effect
 - 1: Clear
- (R/W)

Continued on the next page...

Register 62.1. ADC_CTRL_REG (0x0000)

Continued from the previous page...

ADC_XPD_SAR1_FORCE Keep HP ADC1 powered on forcefully.

0: No effect

1: Power on forcefully

(R/W)

ADC_XPD_SAR2_FORCE Keep HP ADC2 powered on forcefully.

0: No effect

1: Power on forcefully

(R/W)

Register 62.2. ADC_CTRL2_REG (0x0004)

(reserved)								ADC_TIMER_EN				ADC_TIMER_TARGET				(reserved)				ADC_SAR2_INV ADC_SAR1_INV				ADC_MAX_MEAS_NUM				ADC_MEAS_NUM_LIMIT					
31								25				24	23				12				11	10	9	8				1				0	
0 0 0 0 0 0 0 0								0				10				0				0	0	255				0				Reset			

ADC_MEAS_NUM_LIMIT Configures whether to enable the limitation of HP ADC's maximum conversion times.

0: Disable

1: Enable

(R/W)

ADC_MAX_MEAS_NUM Configures the HP ADC's maximum conversion times. (R/W)**ADC_SAR1_INV** Configures whether to invert the data of HP ADC1.

0: Do not invert.

1: Invert the date.

(R/W)

ADC_SAR2_INV Configures whether to to invert the data of HP ADC2.

0: Do not invert.

1: Invert the date.

(R/W)

ADC_TIMER_TARGET Configures HP ADC timer target. (R/W)**ADC_TIMER_EN** Configures whether to enable HP ADC timer trigger.

0: Disable

1: Enable

(R/W)

ADC_FILTER_FACTOR0
ADC_FILTER_FACTOR1
(reserved)

Register 62.6. ADC_SAR1_PATT_TAB3_REG (0x0020)

(reserved)								ADC_SAR1_PATT_TAB3																								
31								24	23																							0
0	0	0	0	0	0	0	0	0x0000																							Reset	

ADC_SAR1_PATT_TAB3 Configures entries 8 ~ 11 for HP ADC FSM1. Each entry is 6-bit wide. See Section [62.5.6 HP ADC Pattern Table](#). (R/W)

Register 62.7. ADC_SAR1_PATT_TAB4_REG (0x0024)

(reserved)								ADC_SAR1_PATT_TAB4																																
31								24								23																							0	
0								0								0								0								0x0000								Reset

ADC_SAR1_PATT_TAB4 Configures entries 12 ~ 15 for HP ADC FSM1. Each entry is 6-bit wide. See Section [62.5.6 HP ADC Pattern Table](#). (R/W)

Register 62.8. ADC_SAR2_PATT_TAB1_REG (0x0028)

(reserved)								ADC_SAR2_PATT_TAB1																															
31								24								23																							0
0 0 0 0 0 0 0 0								0x0000																							Reset								

ADC_SAR2_PATT_TAB1 Configures entries 0 ~ 3 for HP ADC FSM2. Each entry is 6-bit wide. See Section [62.5.6 HP ADC Pattern Table](#). (R/W)

(reserved)

ADC_SAR2_PATT_TAB2

ADC_SAR2_PATT_TAB2 Configures entries 4 ~ 7 for HP ADC FSM2. Each entry is 6-bit wide. See Section [62.5.6 HP ADC Pattern Table](#). (R/W)

(reserved)

ADC_SAR2_PATT_TAB3

ADC_SAR2_PATT_TAB3 Configures entries 8 ~ 11 for HP ADC FSM2. Each entry is 6-bit wide. See Section [62.5.6 HP ADC Pattern Table](#). (R/W)

(reserved)

ADC_SAR2_PATT_TAB4

ADC_SAR2_PATT_TAB4 Configures entries 12 ~ 15 for HP ADC FSM2. Each entry is 6-bit wide. See Section [62.5.6 HP ADC Pattern Table](#). (R/W)

Register 62.12. ADC_FILTER_CTRL0_REG (0x003C)

ADC_FILTER_RESET				(reserved)				ADC_FILTER_CHANNEL0				ADC_FILTER_CHANNEL1				(reserved)			
31	30		24	23		19	18		14	13									0
0	0	0	0	0	0	0	0	0xd		0xd		0	0	0	0	0	0	0	0

Reset

ADC_FILTER_CHANNEL0 Configures the ADC channel for filter 0. Value 0 ~ 7 corresponds to channel 0 ~ 7 in HP ADC1, value 10 ~ 15 corresponds to channel 0 ~ 5 in HP ADC2. (R/W)

ADC_FILTER_CHANNEL1 Configure the ADC channel for filter 1. Same as above. (R/W)

ADC_FILTER_RESET Configures whether to reset the filter.

0: No effect

1: Reset

(R/W)

Register 62.13. ADC_THRESO_CTRL_REG (0x0044)

(reserved)				ADC_THRESO_LOW				ADC_THRESO_HIGH				ADC_THRESO_CHANNEL			
31	30					18	17				5	4			0
0				0						0x1fff				13	

Reset

ADC_THRESO_CHANNEL Configures the ADC channel for threshold monitor 0. Value 0 ~ 7 corresponds to channel 0 ~ 7 in HP ADC1, value 10 ~ 15 corresponds to channel 0 ~ 5 in HP ADC2. (R/W)

ADC_THRESO_HIGH Configures the high threshold for HP ADC threshold monitor 0. (R/W)

ADC_THRESO_LOW Configures the low threshold for HP ADC threshold monitor 0. (R/W)

Register 62.14. ADC_THRES1_CTRL_REG (0x0048)

(reserved)		ADC_THRES1_LOW																ADC_THRES1_HIGH																ADC_THRES1_CHANNEL											
31	30																	18	17																	5	4								0
0	0																0x1fff																13							Reset					

Reset

ADC_THRES1_CHANNEL Configures the ADC channel for threshold monitor 1. Value 0 ~ 7 corresponds to channel 0 ~ 7 in HP ADC1, value 10 ~ 15 corresponds to channel 0 ~ 5 in HP ADC2. (R/W)

ADC_THRES1_HIGH Configures the high threshold for HP ADC threshold monitor 1. (R/W)

ADC_THRES1_LOW Configures the low threshold for HP ADC threshold monitor 1. (R/W)

Register 62.15. ADC_THRES_CTRL_REG (0x004C)

ADC_THRES0_EN						ADC_THRES1_EN						(reserved)						ADC_THRES_ALL_EN						(reserved)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
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0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

ADC_THRES_ALL_EN Configures whether to enable the threshold monitoring for all configured channels.
 0: Disable
 1: Enable
 (R/W)

ADC_THRES1_EN Configures whether to enable threshold monitor 1.
 0: Disable
 1: Enable
 (R/W)

ADC_THRES0_EN Configures whether to enable threshold monitor 0.
 0: Disable
 1: Enable
 (R/W)

Register 62.16. ADC_INT_ENA_REG (0x0050)

ADC_SAR1_DONE_INT_ENA ADC_SAR2_DONE_INT_ENA ADC_THRES0_HIGH_INT_ENA ADC_THRES1_HIGH_INT_ENA ADC_THRES0_LOW_INT_ENA ADC_THRES1_LOW_INT_ENA (reserved)																																	
31	30	29	28	27	26	25																									0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

ADC_THRES1_LOW_INT_ENA Enable bit of [ADC_THRES1_LOW_INT](#). (R/W)

ADC_THRES0_LOW_INT_ENA Enable bit of [ADC_THRES0_LOW_INT](#). (R/W)

ADC_THRES1_HIGH_INT_ENA Enable bit of [ADC_THRES1_HIGH_INT](#). (R/W)

ADC_THRES0_HIGH_INT_ENA Enable bit of [ADC_THRES0_HIGH_INT](#). (R/W)

ADC_SAR2_DONE_INT_ENA Enable bit of [ADC_SAR2_DONE_INT](#). (R/W)

ADC_SAR1_DONE_INT_ENA Enable bit of [ADC_SAR1_DONE_INT](#). (R/W)

Register 62.17. ADC_INT_RAW_REG (0x0054)

ADC_SAR1_DONE_INT_RAW ADC_SAR2_DONE_INT_RAW ADC_THRES0_HIGH_INT_RAW ADC_THRES1_HIGH_INT_RAW ADC_THRES0_LOW_INT_RAW ADC_THRES1_LOW_INT_RAW																																(reserved)																0
31	30	29	28	27	26	25																										0																
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset															

ADC_THRES1_LOW_INT_RAW Raw bit of [ADC_THRES1_LOW_INT](#). (R/WTC/SS)

ADC_THRES0_LOW_INT_RAW Raw bit of [ADC_THRES0_LOW_INT](#). (R/WTC/SS)

ADC_THRES1_HIGH_INT_RAW Raw bit of [ADC_THRES1_HIGH_INT](#). (R/WTC/SS)

ADC_THRES0_HIGH_INT_RAW Raw bit of [ADC_THRES0_HIGH_INT](#). (R/WTC/SS)

ADC_SAR2_DONE_INT_RAW Raw bit of [ADC_SAR2_DONE_INT](#). (R/WTC/SS)

ADC_SAR1_DONE_INT_RAW Raw bit of [ADC_SAR1_DONE_INT](#). (R/WTC/SS)

Register 62.18. ADC_INT_ST_REG (0x0058)

ADC_APB_SARADC1_DONE_INT_ST ADC_APB_SARADC2_DONE_INT_ST ADC_THRES0_HIGH_INT_ST ADC_THRES1_HIGH_INT_ST ADC_THRES0_LOW_INT_ST ADC_THRES1_LOW_INT_ST						(reserved)																														0
31	30	29	28	27	26	25																														0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

ADC_THRES1_LOW_INT_ST Status bit of [ADC_THRES1_LOW_INT](#). (RO)

ADC_THRES0_LOW_INT_ST Status bit of [ADC_THRES0_LOW_INT](#). (RO)

ADC_THRES1_HIGH_INT_ST Status bit of [ADC_THRES1_HIGH_INT](#). (RO)

ADC_THRES0_HIGH_INT_ST Status bit of [ADC_THRES0_HIGH_INT](#). (RO)

ADC_APB_SARADC2_DONE_INT_ST Status bit of [ADC_SAR2_DONE_INT](#). (RO)

ADC_APB_SARADC1_DONE_INT_ST Status bit of [ADC_SAR1_DONE_INT](#). (RO)

Register 62.19. ADC_INT_CLR_REG (0x005C)

ADC_APB_SARADC1_DONE_INT_CLR ADC_APB_SARADC2_DONE_INT_CLR ADC_THRES0_HIGH_INT_CLR ADC_THRES1_HIGH_INT_CLR ADC_THRES0_LOW_INT_CLR ADC_THRES1_LOW_INT_CLR						(reserved)																													0
31	30	29	28	27	26	25																													0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

ADC_THRES1_LOW_INT_CLR Clear bit of [ADC_THRES1_LOW_INT](#). (WT)

ADC_THRES0_LOW_INT_CLR Clear bit of [ADC_THRES0_LOW_INT](#). (WT)

ADC_THRES1_HIGH_INT_CLR Clear bit of [ADC_THRES1_HIGH_INT](#). (WT)

ADC_THRES0_HIGH_INT_CLR Clear bit of [ADC_THRES0_HIGH_INT](#). (WT)

ADC_APB_SARADC2_DONE_INT_CLR Clear bit of [ADC_SAR2_DONE_INT](#). (WT)

ADC_APB_SARADC1_DONE_INT_CLR Clear bit of [ADC_SAR1_DONE_INT](#). (WT)

ADC_APB_ADC_TRANS
ADC_APB_ADC_RESET_FSM

(reserved)

ADC_APB_ADC_EOF_NUM

(R/W)

(R/W)

(reserved)

ADC_CTRL_DATE

ADC_CTRL_DATE HP ADC version control register. (RO)

The addresses in the following section are relative to LP ADC's base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

ESP32-P4 TRM
PRELIMINARY

LPADC_SAR1_EN_PAD_FORCE_ENABLE
LPADC_SAR1_INT_EN
LPADC_SAR1_DATA_INV

(reserved)

LPADC_SAR1_CLK_DIV

Reset

(R/W)

(R/W)

Register 62.23. LPADC_MEAS1_CTRL2_REG (0x000C)

LPADC_SAR1_EN_PAD_FORCE											LPADC_SAR1_EN_PAD											LPADC_MEAS1_START_FORCE											LPADC_MEAS1_START_SAR											LPADC_MEAS1_DONE_SAR											LPADC_MEAS1_DATA_SAR										
31	30																		19	18	17	16	15																		0																								
0	0x0																		0	0	0	0x00																		Reset																									

LPADC_MEAS1_DATA_SAR Stores LP ADC1's conversion data in one-shot sampling mode. (RO)

LPADC_MEAS1_DONE_SAR Indicates whether LP ADC1 conversion is done. (RO)

LPADC_MEAS1_START_SAR Configures whether to start LP ADC1 by software.

0: No effect

1: Start LP ADC1 by software

Valid only when [LPADC_MEAS1_START_FORCE](#) = 1.

(R/W)

LPADC_MEAS1_START_FORCE Configures whether to use software to enable LP ADC1 sampling.

0: Select FSM to start LP ADC1 sampling

1: Select software to start LP ADC1 sampling

(R/W)

LPADC_SAR1_EN_PAD Configures the sampling channel for LP ADC1. Value 0 ~ 7 corresponds to channel 0 ~ 7. (R/W)

LPADC_SAR1_EN_PAD_FORCE Configures whether to use software to select the sampling channel for LP ADC1.

0: No effect

1: Use software to control

(R/W)

LPADC_SAR1_DIG_FORCE

(reserved)

Reset

(R/W)

LPADC_SAR1_ATTEN

Reset

(R/W)

(reserved)
LPADC_SAR2_INT_EN
LPADC_SAR2_DATA_INV
LPADC_SAR2_EN_PAD_FORCE_ENABLE

(reserved)

LPADC_SAR2_CLK_DIV

Reset

(R/W)

(R/W)

ESP32-P4 TRM
PRELIMINARY

Register 62.27. LPADC_MEAS2_CTRL2_REG (0x0030)

LPADC_SAR2_EN_PAD_FORCE											LPADC_SAR2_EN_PAD											LPADC_MEAS2_START_FORCE											LPADC_MEAS2_START_SAR											LPADC_MEAS2_DONE_SAR											LPADC_MEAS2_DATA_SAR										
31	30																		19	18	17	16	15																	0																									
0	0x0																		0	0	0	0x00																	Reset																										

LPADC_MEAS2_DATA_SAR Stores LP ADC2's conversion data in one-shot sampling mode. (RO)

LPADC_MEAS2_DONE_SAR Indicates whether LP ADC2 conversion is done. (RO)

LPADC_MEAS2_START_SAR Configures whether to start LP ADC2 by software.

0: No effect

1: Start LP ADC2 by software

Valid only when [LPADC_MEAS2_START_FORCE](#) = 1.

(R/W)

LPADC_MEAS2_START_FORCE Configures whether to use software to enable LP ADC2 sampling.

0: Select FSM to start LP ADC2 sampling

1: Select software to start LP ADC2 sampling

(R/W)

LPADC_SAR2_EN_PAD Configures the sampling channel for LP ADC2. Value 2 ~ 7 corresponds to channel 0 ~ 5. (R/W)

LPADC_SAR2_EN_PAD_FORCE Configures whether to use software to select the sampling channel for LP ADC1.

0: No effect

1: Use software to control

(R/W)

Register 62.28. LPADC_MEAS2_MUX_REG (0x0034)

Diagram of the LPADC_SAR2_LP_FORCE register. The register is 32 bits wide. Bit 31 is labeled 'LPADC_SAR2_LP_FORCE'. Bits 30 to 0 are labeled '(reserved)'. A 'Reset' value of 0 is indicated for all bits.

LPADC_SAR2_LP_FORCE Configures whether to disable HP ADC2 Controller to control SAR ADC2.

0: Do not disable HP ADC2 Controller.

1: Disable HP ADC2 Controller.

(R/W)

Register 62.29. LPADC_ATTEN2_REG (0x0038)

Diagram of the LPADC_SAR2_ATTEN register. The register is 32 bits wide, with bit positions 31 down to 0 labeled. The value 0xffffffff is shown, indicating all bits are set to 1. The label "Reset" is at the bottom right.

LPADC_SAR2_ATTEN Configures the attenuation value for one-shot sampling of LP ADC2.

0: 0 dB

1: 2.5 dB

2: 6 dB

3: 12 dB

(R/W)

LPADC_FORCE_XPD_SAR2
LPADC_FORCE_XPD_SAR1

(reserved)

LPADC_FOR LPADC_FOR

- Reset

LPADC_FOR LPADC_FOR

- 3: Software control, force on.
2: Software control, force off.
1/0: Hardware control.
(R/W)

Register 62.31. LPADC_COCPU_INT_RAW_REG (0x0048)

Diagram of the LPADC_COCPU_SARADC1_WAKE_INT_RAW register. The register is 32 bits wide. Bits 31-6 are reserved. Bits 5-0 are labeled LPADC_COCPU_SARADC1_WAKE_INT_RAW. The register is reset to 0.

LPADC_COCPU_SARADC1_INT_RAW Raw bit of [LPADC_COCPU_SARADC1_DONE_INT](#). (R/WTC/SS)

LPADC_COCPU_SARADC2_INT_RAW Raw bit of LPADC_COCPU_SARADC2_DONE_INT.
(R/WTC/SS)

LPADC_COCPU_SARADC1_ERROR_INT_RAW Raw bit of LPADC_COCPU_SARADC1_ERROR_INT.
(R/WTC/SS)

LPADC_COCPU_SARADC2_ERROR_INT_RAW Raw bit of LPADC_COCPU_SARADC2_ERROR_INT.
(R/WTC/SS)

LPADC_COCPU_SARADC1_WAKE_INT_RAW Raw bit of LPADC_COCPU_SARADC1_WAKE_INT.
(R/WTC/SS)

LPADC_COCPU_SARADC2_WAKE_INT_RAW Raw bit of LPADC_COCPU_SARADC2_WAKE_INT.
(R/WTC/SS)

Register 62.32. LPADC_INT_ENA_REG (0x004C)

(reserved)																LPADC_COCPU_SARADC2_WAKE_INT_ENA			
																LPADC_COCPU_SARADC1_WAKE_INT_ENA			
																LPADC_COCPU_SARADC2_ERROR_INT_ENA			
																LPADC_COCPU_SARADC1_ERROR_INT_ENA			
																LPADC_COCPU_SARADC2_INT_ENA			
																LPADC_COCPU_SARADC1_INT_ENA			
31																6	5	4	3
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
																2	1	0	Reset
																0	0	0	0

LPADC_COCPU_SARADC1_INT_ENA Enable bit of LPADC_COCPU_SARADC1_DONE_INT. (R/WTC)

LPADC_COCPU_SARADC2_INT_ENA Enable bit of LPADC_COCPU_SARADC2_DONE_INT. (R/WTC)

LPADC_COCPU_SARADC1_ERROR_INT_ENA Enable bit of LPADC_COCPU_SARADC1_ERROR_INT.
(R/WTC)

LPADC_COCPU_SARADC2_ERROR_INT_ENA Enable bit of LPADC_COCPU_SARADC2_ERROR_INT.
(R/WTC)

LPADC_COCPU_SARADC1_WAKE_INT_ENA Enable bit of LPADC_COCPU_SARADC1_WAKE_INT.
(R/WTC)

LPADC_COCPU_SARADC2_WAKE_INT_ENA Enable bit of LPADC_COCPU_SARADC2_WAKE_INT.
(R/WTC)

Register 62.33. LPADC_INT_ST_REG (0x0050)

(reserved)

31 6 5 4 3 2 1 0

LPADC_COCPU_SARADC2_WAKE_INT_ST
 LPADC_COCPU_SARADC1_WAKE_INT_ST
 LPADC_COCPU_SARADC2_ERROR_INT_ST
 LPADC_COCPU_SARADC1_ERROR_INT_ST
 LPADC_COCPU_SARADC2_INT_ST
 LPADC_COCPU_SARADC1_INT_ST

Reset

LPADC_COCPU_SARADC1_INT_ST Status bit of LPADC_COCPU_SARADC1_DONE_INT. (RO)

LPADC_COCPU_SARADC2_INT_ST Status bit of LPADC_COCPU_SARADC2_DONE_INT. (RO)

LPADC_COCPU_SARADC1_ERROR_INT_ST Status bit of LPADC_COCPU_SARADC1_ERROR_INT.
(RO)

LPADC_COCPU_SARADC2_ERROR_INT_ST Status bit of LPADC_COCPU_SARADC2_ERROR_INT.
(RO)

LPADC_COCPU_SARADC1_WAKE_INT_ST Status bit of LPADC_COCPU_SARADC1_WAKE_INT. (RO)

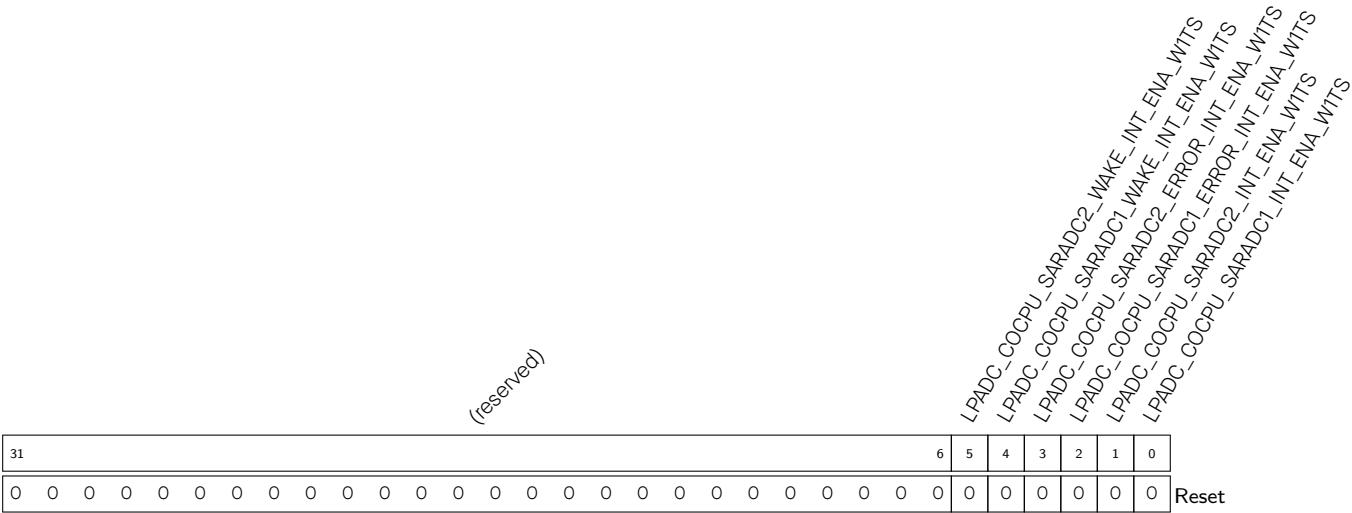
LPADC_COCPU_SARADC2_WAKE_INT_ST Status bit of LPADC_COCPU_SARADC2_WAKE_INT.
(RO)

Register 62.34. LPADC_INT_CLR_REG (0x0054)

(reserved)																												LPADC_COCPU_SARADC2_WAKE_INT_CLR LPADC_COCPU_SARADC1_WAKE_INT_CLR LPADC_COCPU_SARADC2_ERROR_INT_CLR LPADC_COCPU_SARADC1_ERROR_INT_CLR LPADC_COCPU_SARADC2_INT_CLR LPADC_COCPU_SARADC1_INT_CLR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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- LPADC_COCPU_SARADC1_INT_CLR Clear bit of LPADC_COCPU_SARADC1_DONE_INT. (WT)
- LPADC_COCPU_SARADC2_INT_CLR Clear bit of LPADC_COCPU_SARADC2_DONE_INT. (WT)
- LPADC_COCPU_SARADC1_ERROR_INT_CLR Clear bit of LPADC_COCPU_SARADC1_ERROR_INT.
(WT)
- LPADC_COCPU_SARADC2_ERROR_INT_CLR Clear bit of LPADC_COCPU_SARADC2_ERROR_INT.
(WT)
- LPADC_COCPU_SARADC1_WAKE_INT_CLR Clear bit of LPADC_COCPU_SARADC1_WAKE_INT. (WT)
- LPADC_COCPU_SARADC2_WAKE_INT_CLR Clear bit of LPADC_COCPU_SARADC2_WAKE_INT.
(WT)

Register 62.35. LPADC_INT_ENA_W1TS_REG (0x0058)



LPADC_COCPU_SARADC1_INT_ENA_W1TS Write 1 to enable LPADC_COCPU_SARADC1_INT_ENA.
(WT)

LPADC_COCPU_SARADC2_INT_ENA_W1TS Write 1 to enable LPADC_COCPU_SARADC2_INT_ENA.
(WT)

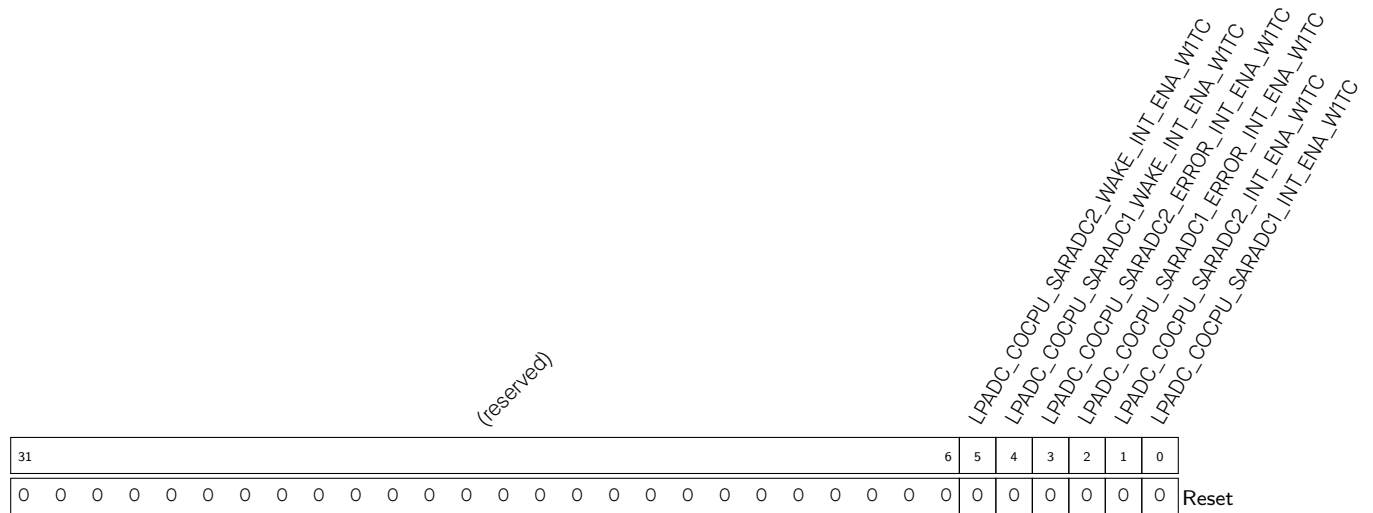
LPADC_COCPU_SARADC1_ERROR_INT_ENA_W1TS Write 1 to enable LPADC_COCPU_SARADC1_ERROR_INT_ENA. (WT)

LPADC_COCPU_SARADC2_ERROR_INT_ENA_W1TS Write 1 to enable LPADC_COCPU_SARADC2_ERROR_INT_ENA. (WT)

LPADC_COCPU_SARADC1_WAKE_INT_ENA_W1TS Write 1 to enable LPADC_COCPU_SARADC1_WAKE_INT_ENA. (WT)

LPADC_COCPU_SARADC2_WAKE_INT_ENA_W1TS Write 1 to enable LPADC_COCPU_SARADC2_WAKE_INT_ENA. (WT)

Register 62.36. LPADC_INT_ENA_W1TC_REG (0x005C)



LPADC_COCPU_SARADC1_INT_ENA_W1TC Write 1 to clear LPADC_COCPU_SARADC1_INT_ENA.
(WT)

LPADC_COCPU_SARADC2_INT_ENA_W1TC Write 1 to clear LPADC_COCPU_SARADC2_INT_ENA.
(WT)

LPADC_COCPU_SARADC1_ERROR_INT_ENA_W1TC Write 1 to clear
LPADC_COCPU_SARADC1_ERROR_INT_ENA. (WT)

LPADC_COCPU_SARADC2_ERROR_INT_ENA_W1TC Write 1 to clear
LPADC_COCPU_SARADC2_ERROR_INT_ENA. (WT)

LPADC_COCPU_SARADC1_WAKE_INT_ENA_W1TC Write 1 to clear
LPADC_COCPU_SARADC1_WAKE_INT_ENA. (WT)

LPADC_COCPU_SARADC2_WAKE_INT_ENA_W1TC Write 1 to clear
LPADC_COCPU_SARADC2_WAKE_INT_ENA. (WT)

Register 62.37. LPADC_WAKEUP1_REG (0x0060)

LPADC_SAR1_WAKEUP_MODE						LPADC_SAR1_WAKEUP_EN						LPADC_SAR1_WAKEUP_OVER_UPPER_TH						LPADC_SAR1_WAKEUP_TH_HIGH						LPADC_SAR1_WAKEUP_TH_LOW					
(reserved)												(reserved)																	
31	30	29	28	26	25	14	13	12	11	0																			
0	0	0	0	0	0	0xfff						0	0	0						Reset									

LPADC_SAR1_WAKEUP_TH_LOW Configures the low threshold for LP ADC1's wake-up function.
(R/W)

LPADC_SAR1_WAKEUP_TH_HIGH Configures the high threshold for LP ADC1's wake-up function.
(R/W)

LPADC_SAR1_WAKEUP_OVER_UPPER_TH Indicates whether the wake-up event is triggered by the conversion data exceeding high threshold. (RO)

LPADC_SAR1_WAKEUP_EN Configures whether to enable wake-up function for LP ADC1.
0: Disable
1: Enable
(R/W)

LPADC_SAR1_WAKEUP_MODE Selects the wake-up mode for automatic monitoring. Valid only when [LPADC_SAR1_WAKEUP_EN](#)=1.
0: Absolute value mode
1: Change value mode
(R/W)

Register 62.38. LPADC_WAKEUP2_REG (0x0064)

LPADC_SAR2_WAKEUP_MODE						LPADC_SAR2_WAKEUP_EN						LPADC_SAR2_WAKEUP_OVER_UPPER_TH						LPADC_SAR2_WAKEUP_TH_HIGH						LPADC_SAR2_WAKEUP_TH_LOW					
(reserved)												(reserved)																	
31	30	29	28	26	25	14	13	12	11	0																			
0	0	0	0	0	0	0xfff						0	0	0						Reset									

LPADC_SAR2_WAKEUP_TH_LOW Configures the low threshold for LP ADC2's wake-up function.
(R/W)

LPADC_SAR2_WAKEUP_TH_HIGH Configures the high threshold for LP ADC2's wake-up function.
(R/W)

LPADC_SAR2_WAKEUP_OVER_UPPER_TH Indicates whether the wake-up event is triggered by the conversion data exceeding high threshold. (RO)

LPADC_SAR2_WAKEUP_EN Configures whether to enable wake-up function for LP ADC2.
0: Disable
1: Enable
(R/W)

LPADC_SAR2_WAKEUP_MODE Selects the wake-up mode for automatic monitoring. Valid only when [LPADC_SAR2_WAKEUP_EN](#)=1.
0: Absolute value mode
1: Change value mode
(R/W)

Register 62.39. LPADC_WAKEUP_SEL_REG (0x0068)

(reserved)																										LPADC_SAR_WAKEUP_SEL	
31																										1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

LPADC_SAR_WAKEUP_SEL Enables the wake-up function for LP ADC1 or LP ADC2. 0: LP ADC1.
1: LP ADC2.
(R/W)

Register 62.40. LPADC_SAR1_HW_WAKEUP_REG (0x006C)

(reserved)																LPADC_ADC1_HW_READ_RATE_I																LPADC_ADC1_HW_READ_EN_I																																															
31																17																16																1																0															
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0x64																0																Reset																															

LPADC_ADC1_HW_READ_EN_I Configures whether to enable automatic monitoring for LP ADC1.

0: Disable

1: Enable

(R/W)

LPADC_ADC1_HW_READ_RATE_I Configures the sampling rate for hardware-triggered automatic monitoring for LP ADC1. The sampling period = configured value × LP ADC1's working clock cycle. (R/W)

Register 62.41. LPADC_SAR2_HW_WAKEUP_REG (0x0070)

Diagram illustrating the structure of the LPADC_ADC2_HW_READ_EN register:

- Bits 31 to 17: (reserved)
- Bits 16 to 1: LPADC_ADC2_HW_READ_RATE_I
- Bit 0: LPADC_ADC2_HW_READ_EN_I

The register is divided into three sections: a 16-bit reserved field, a 16-bit field for LPADC_ADC2_HW_READ_RATE_I, and a 1-bit field for LPADC_ADC2_HW_READ_EN_I.

LPADC_ADC2_HW_READ_EN_I Configures whether to enable automatic monitoring for LP ADC2.

0: Disable

1: Enable

(R/W)

LPADC_ADC2_HW_READ_RATE_I Configures the sampling rate for hardware-triggered automatic monitoring for LP ADC2. The sampling period = configured value × LP ADC2's working clock cycle. (R/W)